Refined error estimates and decay properties for a damped semilinear wave equation

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This talk will concern the approximation of a semilinear wave equation with space-dependent damping in one space dimension. After rewriting the equation as a first order system, we present an approach for proving rigorous L^1 error estimates for certain classes of approximations. The main relevant features are that these approximations preserve stationary solutions and that the L^1 difference with exact solutions is bounded uniformly in time, therefore leading to accurate estimates for large times. Moreover, a decay property of the total variation of the solution is shown in the region where the damping is supported. This will be related to a well-known property of decay of energy for the damped wave equation.

Joint work with L. Gosse (IAC–CNR, Rome, Italy).

[1] D. Amadori, L. Gosse. Error Estimates for Well-Balanced and Time-Split Schemes on a Damped Semilinear Wave Equation. *Math. Comp.* **85** (2016), 601-633

[2] D. Amadori, L. Gosse. Error Estimates for Well-Balanced Schemes on Simple Balance Laws. One-Dimensional Position-Dependent Models, *SpringerBriefs in Mathematics*, 2015